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ORIGINAL RESEARCH

Platelet-derived Growth Factor-α and Neuropilin-1 Mediate Lung Fibroblast Response to Rigid Collagen Fibers

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Abstract

During pulmonary secondary alveolar septation, the rudimentary distal saccule subdivides by extending tissue sheets into the saccular air space, creating alveoli, which open into the alveolar duct. The sheets originate from saccular mesenchymal cells, which contain α-SMA (αSMA [ACTA2]) and abut elastic fibers (myofibroblasts [MF]), characteristics that are shared by cells that subsequently occupy the secondary septal tips. During elongation, collagen fibers are positioned to provide a scaffold for translocating septal mesenchymal cells. We hypothesized that collagen fibers direct the migration, orientation, and location of MFs during septal elongation. To address this hypothesis, we examined how electrospun collagen fibers direct the migration of fibroblasts bearing targeted deletions of PDGFRα (platelet-derived growth factor receptor-α) or Nrp1 (neuropilin-1), after their isolation from lungs that exhibit reduced secondary septation. We observed that deletion of either gene reduced Rac1 activation and the speed of migration of lung fibroblasts (LF) along electrospun fibers. The deletions did not reduce the proportion of LF that displayed collagen-binding integrins and increased the proportion of LF bearing activated β1-integrin. LF bearing the PDGFRα deletion failed to localize focal adhesions over electrospun fibers, suggesting that they may not appropriately sense and respond to regionally increased stiffness near the fibers. In lungs of mice bearing the PDGFRα deletion, collagen fibers are delocalized from ACTA2-containing MF, and their orientation deviated from the plane of the alveolar walls. Diminished PDGFRα or Nrp1 reduces LF localization to stiffer regions of fibrillar collagen substrates, suggesting that signaling through these receptors enables responsiveness to regional differences in extracellular matrix rigidity.

Keywords: alveolar development; mechanotransduction; myofibroblast; cell migration

At birth, the terminal saccules of mammalian lungs abruptly transition from prenatal distension by dampened fluid pulsations to much larger swings in volume during gaseous respiration. This alters the stimuli that guide cell proliferation, differentiation, and positioning. Greater air space expansion increases the mechanical forces contacting interstitial cells within the rudimentary air sacs as they transition to alveoli (alveolarization). Alveolar fibroblasts play a key role during this transition by structurally supporting the air–blood interface as it expands nearly fourfold during the first 2 postnatal weeks in mice (1). Alveolar fibroblasts (here referred to as lung fibroblasts, LF) must proliferate, migrate, and produce structural extracellular matrix proteins. This is balanced with their role as contractile cells, which adjust the cytoskeleton and surrounding matrix to rapid fluctuations in volume. Alveolar fibroblasts, which contain ACTA2 (αSMA) and are often referred to

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Author Contributions: S.E.M. conceived the studies, performed some experiments, analyzed data, and wrote the manuscript. T.I.L. performed the studies using the atomic force microscope and, along with A.V.T., analyzed the data and edited the manuscript. D.M.M. performed experiments, analyzed data, and edited the manuscript. L.Z. performed experiments and edited the manuscript.

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Clinical Relevance

Although it is widely held that regional mechanics influence cellular behavior, the responsible mechanisms remain largely unidentified. This study shows how two signaling proteins regulate the way lung fibroblasts interact with and migrate along collagen fibers and how lung structure is altered in the absence of these interactions.

as myofibroblasts (MF), are most abundant where the alveolar ducts exit to alveoli (alveolar entry rings, AER), and where respiration-induced tensile forces are high (2). We hypothesize that higher mechanical tension at the AER regulates the migration of MF as secondary septa subdivide the rudimentary saccules.

It remains unclear how mechanical tension at the AER alters the behavior of MF. We posited that collagen fibers, which are the first load-bearing fibers to populate the septa, form a scaffold along which LF migrate and ensure optimal positioning of structural proteins (3, 4). This would stabilize the more fragile epithelial and endothelial cells and ensure their propinquity. As secondary septa form, MF deposit elastic fibers, which during subsequent maturation form the core, whereas collagen wraps the continuous fiber network, which connects alveoli to the visceral pleura (5, 6).

Cell surface integrins (primarily $\alpha_1\beta_1$ and $\alpha_2\beta_1$) bind fibrillar collagens, of which collagens I and III are the most abundant within the alveolar interstitium (7). When contacting extracellular collagen fibers, integrins cluster and connect to the actin cytoskeleton through focal adhesions (FAs) via adapters such as talin and kindlin (8). The integrin–actin nexus is strengthened by recruitment of vinculin, actinin, paxillin, and zyxin, which promote actin bundling. As actin is stabilized, Myo2 (myosin 2) is recruited and, through contraction, repositions the cell in response to the fibrillar extracellular matrix (9). Rho-GTPase is an important mediator of Myo2 motor activity by switching on ROCK (Rho-activated kinase) and by increasing actin polymerization (10). However, cell migration requires FA turnover, particularly at the leading edge where Rac1 regulates branched-actin formation and FAK (focal adhesion kinase) regulates FA turnover (11). PDGFRα (platelet-derived growth factor receptor-α) and other tyrosine kinases regulate cell migration by controlling lamellipodia formation and FA turnover through Rac-GTPase and FAK (12, 13).

PDGF-A (platelet-derived growth factor-A) and its sole receptor, PDGFRα, are absolutely required for alveolar septation; in their absence MF fail to proliferate, and the elastic fiber network and secondary septa do not form (14). Signaling by PDGF-A through PDGFRα is required for MF proliferation and differentiation, and when signaling is disrupted, cultured LF exhibit reduced durotaxis (migration toward a stiffer substrate) (15).

Nrp1 (neuropilin-1) is a transmembrane signaling protein that lacks intrinsic kinase activity but enhances the signaling of other PTK (protein tyrosine kinases), such as VEGF (vascular endothelial growth factor), PDGFRβ, and PDGFRα (16). This increases cell migration, integrin signaling, FA remodeling, and neuronal guidance cues (17, 18). Neuropilin indirectly regulates cellular interactions with collagen through its effects on integrins. Nrp1 promotes αVβ3, α2β1, and α5β1 integrin recycling and FA turnover to increase migration of mesangial cells and pericytes (17, 19). Acting through αV and α5 integrins, cancer-associated fibroblasts contract surrounding collagen and increase the migration of cocultured epithelial cancer cells (20). In this study, we examine how PDGFRα and Nrp1 enhance LF migration along collagen fibers during alveolarization.

Methods

Mice

Pdgfrα$^{tm11(EGFP)Sor/J}$ mice have been described elsewhere, and are referred to as PDGFRα-GFP mice (21). Production and nuclear localization of eGFP is under the control of the endogenous *pdgfrα* promoter. GFP expression spatially and temporally recapitulates endogenous *pdgfrα* gene expression. We used heterozygote mice carrying one *pdgfrα*-GFP allele (which does not encode active PDGFRα) and one functional *pdgfrα* allele, which are phenotypically identical to wild-type (GFP^{neg}) mice (22). Mice with *tagln*-Cre–mediated deletion of LoxP-flanked *pdgfrα* or *nrp1* have been described (21). Transgelin (*tagln*) is expressed in pulmonary myofibroblasts, pericytes, and smooth muscle cells. Protocols for animal use were approved by the Iowa City Veterans Affairs Medical Center animal use committee (21). Both males and females were studied.

Isolation of Lung Fibroblasts

For flow cytometry (FACS) at postnatal Day 8 (P8) or for culture (P10 or P12), LF were isolated from mice with the following genotypes: heterozygous PDGFRα-GFP; TGCre$^{+/-}$; PDGFRαF/F, TGCre$^{+/-}$; Nrp1F/F (the heterozygous *tagln*-Cre allele excises LoxP-flanked segment of *pdgfrα* or *nrp1* DNA); and from littermate TGCre$^{+/-}$; PDGFRαF/− or TGCre$^{+/-}$; Nrp1F/− (only one *nrp1* LoxP flanked allele). Mesenchymal cells were isolated from parenchymal endothelial and epithelial cells by differential adherence to tissue culture–treated plastic (23).

Electrospinning of Collagen Fibers and LF Migration

A detailed description appears in the data supplement. The fibers were deposited on glass cover slides with their predominant orientation parallel to the channels used for migration. Fibroblasts were embedded in 1 mg/ml bovine calf skin collagen to form a soft gel surrounding the electrospun fibers in the migration channels.

The procedures involving analysis of Rac1-GTP with a Förster resonance energy transfer (FRET) biosensor (15), Western IB, analytical flow cytometry, and the proximity ligation assay (23) have been published and are described in data supplement. Detailed descriptions of the procedures used to quantify activation of latent-TGFβ in fibroblast cultures, measure the elasticity (Young's moduli) of collagen fibers and gels using atomic force microscopy (AFM), and analyses of images acquired by laser scanning confocal or second harmonic generation (SHG) microscopy appear in the data supplement. Because 1-mg/ml collagen gels were dislodged when mounted on the AFM, we increased the concentration to 2 mg/ml and used 3% glutaraldehyde to bond the gel to glass. Uniform conditions for laser power, detector gain, acquisition speed, and z-slice depth were applied to all samples within an imaging session, which always included

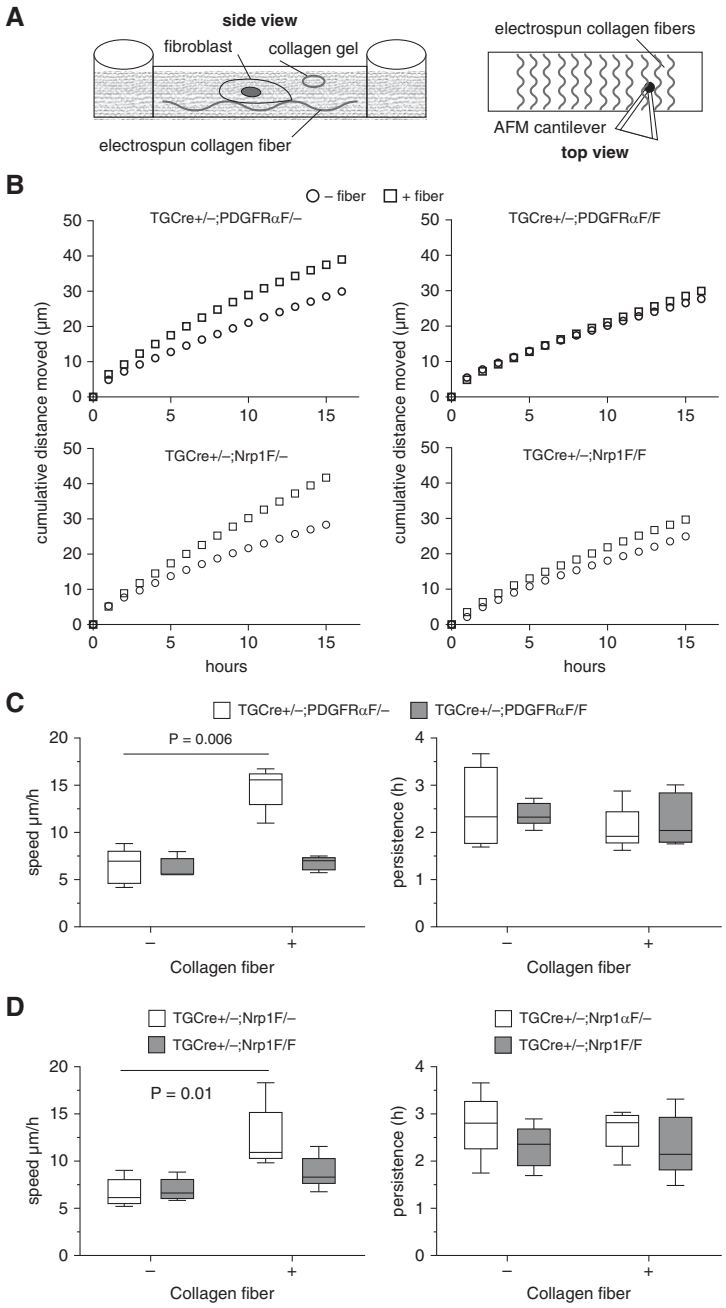

Figure 1. Lung fibroblasts (LF) migrate along electrospun collagen fibers embedded in a three-dimensional collagen gel. Between postnatal Days 10 and 12, LF were isolated (based on differential adherence to plastic; *see* data supplement) from mice bearing one (F/−) or two (F/F) LoxP-flanked (Floxed) alleles for PDGFRα (platelet-derived growth factor receptor-α) or Nrp1 (neuropilin-1) and one allele with Cre-recombinase driven by *tagln1* (TGCre$^{+/-}$). (*A*) By electrospinning, bovine calf skin collagen was deposited on a glass cover slide, which was attached to an Ibidi Sticky-Slide. After loading them with CellTracker-Green fluorescent dye, the LF were mixed with 1 mg/ml collagen and a gel formed within wells (*A*, stippled), encasing the electrospun fibers that resided in some of the wells. Individual cells were tracked for 16 hours using time-lapse imaging. (*B*) Plots showing the cumulative distance migrated by representative LF from TGCre$^{+/-}$; PDGFRα F/− or littermate TGCre$^{+/-}$; PDGFRαF/F; or TGCre$^{+/-}$; Nrp1 F/− and TGCre$^{+/-}$; Nrp1F/F mice. (*C* and *D*) The cell trajectories were analyzed using a random walk model to calculate speed and persistence for TGCre$^{+/-}$; PDGFRα F/− or TGCre$^{+/-}$; PDGFRαF/F (*C*) and TGCre$^{+/-}$; Nrp1 F/− or TGCre$^{+/-}$; Nrp1F/F (*D*). Box-and-whisker (Tukey) plots of data from five separate cell isolations for each genotype, two-way ANOVA. AFM = atomic force microscopy.

lungs from both gene-deleted and -undeleted littermate control mice.

Statistical Methods

Box plots extend from the 25th to the 75th percentiles, and whiskers were plotted based on the Tukey interquartile range using GraphPad Prism. *N* is the number of different mice that were used or the number of different experiments that were performed using cell cultures. ANOVA (either one- or two-way, as indicated in the figure legends) was performed using Systat and Student's *t* test using Microsoft Excel (either paired or unpaired, as indicated in the figure legends). Multiple *t* test was performed using GraphPad Prism 7. Values of $P < 0.05$ were considered significant.

Results

Structural and Mechanical Characteristics of Electrospun Collagen within a Collagen

A schematic diagram of an LF embedded within a soft-fibrillar collagen gel contacting an electrospun collagen fiber is shown in Figure 1A.

AFM Mapping Showed That Electrospun Collagen Fibers Influenced the Regional Stiffness of the Surrounding Collagen Gel

Electrospun fibers imaged in air (*see* Figure E1A in the data supplement) or immersed in PBS without a surrounding collagen gel (Figure E1B) exhibited Young's moduli of 1,000 and 0.1 MPa, respectively. The elasticity in two areas distant (*d*, >50 μm, now shown in Figure E2) from electrospun collagen fibers was 70 ± 40 Pa and 70 ± 30 (mean ± 1 SD). The elasticity values calculated for each force map over the regions proximate (*p*) to underlying electrospun collagen fibers (Figure E2, region 1*p* and region 2*p*) were separated into two sectors: <110 Pa was attributed to the fiber-free gel response, and >110 Pa was attributed to the underlying electrospun fibers. The region *d* and region *p* data sets for two trials were then combined into histograms, which clearly showed bimodal distributions (Figure E3). Each force-plot distribution collected over electrospun fibers was fit to a Gaussian function, yielding the most probable

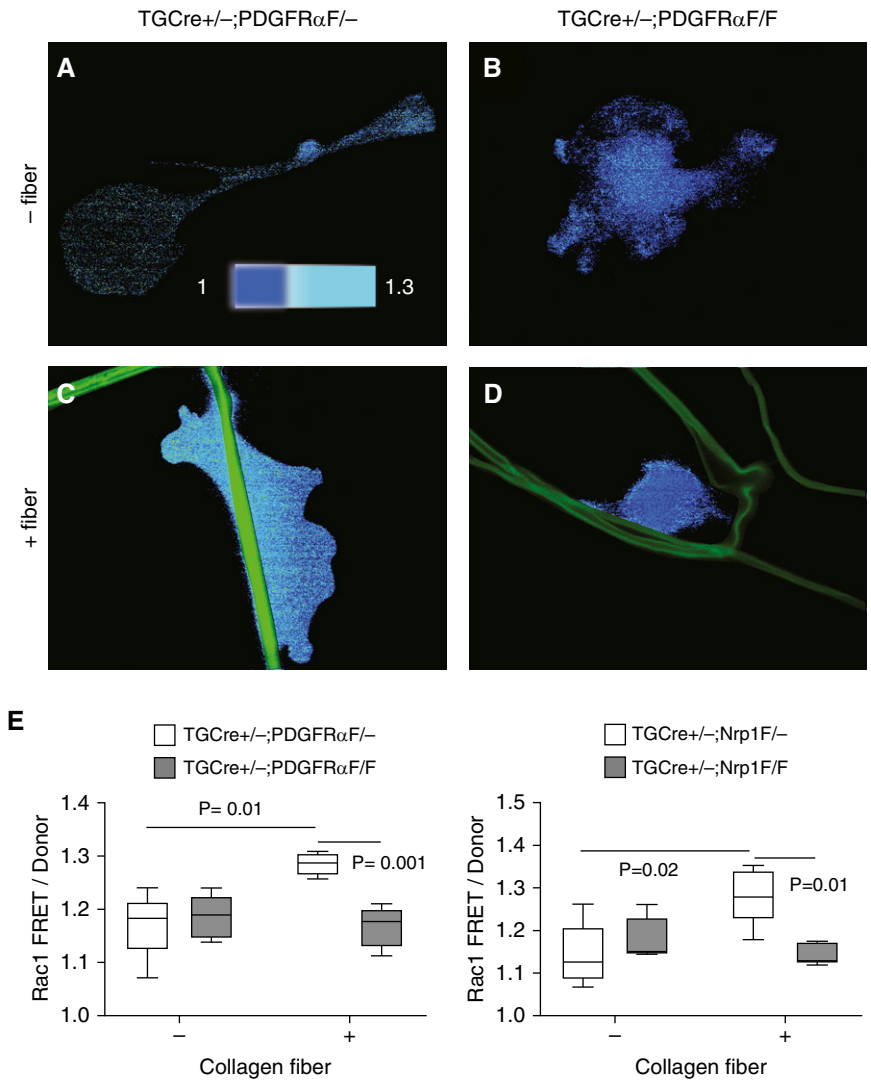

Figure 2. Electrospun collagen fibers increase Rac1-GTP only in LF with a full complement of PDGFRα or Nrp1. (*A*) Fibroblasts were isolated from the lungs of TGCre$^{+/-}$; PDGFRα F/− or littermate TGCre$^{+/-}$; PDGFRαF/F and TGCre$^{+/-}$; Nrp1 F/− or TGCre$^{+/-}$; Nrp1F/F mice. The LF were embedded in a 1 mg/ml collagen gel and loaded into an Ibidi Sticky-Slide containing electrospun collagen fibers. Cells that contacted or did not contact electrospun fibers were imaged, and regions of interest were designated at lamellipodia or adjacent to where LF contacted a fiber. (*A–D*) Ratios of Förster resonance energy transfer (FRET) intensity relative to donor intensity for control (*A* and *C*) and *pdgfrα* gene-deleted LF (*B* and *D*) are shown. (*E*) Box-and-whisker (Tukey) plots of data pooled from five separate cell isolations, pooling ratio FRET (rFRET) from 12 separate cells per condition for each isolation, for TGCre$^{+/-}$; PDGFRα F/− or TGCre$^{+/-}$; PDGFRαF/F and four separate isolations for TGCre$^{+/-}$; Nrp1 F/− or TGCre$^{+/-}$; Nrp1F/F. *y*-axis is the mean pixel by pixel ratio for the intensity of the FRET versus the donor channel (rFRET) within all regions of interest at the periphery, for a particular cell. Box-and-whisker (Tukey) plots, two-way ANOVA.

elasticity value and 1 SD of 190 ± 40 Pa (region 1*p*) and 180 ± 40 Pa (region 2*p*), where the proximate fiber increased elasticity by ∼2.5-fold compared with collagen gel alone. The spatially mapped regions with elasticity >110 Pa visually correspond to the underlying fluorescein-labeled electrospun fibers (Figure E2), corroborating that increased regional elasticity is conferred by the proximate electrospun fibers.

LF Migrated Faster along Electrospun Collagen Fibers

LF were isolated from the lungs of mice bearing transgelin-targeted deletions of PDGFRα (TGCre$^{+/-}$; PDGFRαF/F) or Nrp1 (TGCre$^{+/-}$; Nrp1F/F) or their littermate control mice, which bore only one LoxP-flanked allele (F/−). The LF were embedded in a 1-mg/ml solution of collagen, which formed a soft gel around the more rigid electrospun collagen fibers. Time-lapse images were acquired of LF, which were either associated with collagen fibers or remained distant from electrospun fibers throughout the 16-hour period. The cumulative distance moved (Figure 1B) was higher when control but not PDGFRα- or Nrp1-depleted LF contacted the electrospun fibers. Association with electrospun fibers increased the speed of migration for LF from the lungs of control mice (TGCre$^{+/-}$; PDGFRαF/− or TGCre$^{+/-}$; Nrp1F/−) but not those from their TGCre$^{+/-}$; PDGFRαF/F (Figure 1C) or TGCre$^{+/-}$; Nrp1F/F (Figure 1D) littermates. Electrospun fibers did not alter the directional persistence of migration of LF of any genotype. Therefore, PDGFRα and Nrp1 are required for the stiffer electrospun fibers to increase the speed of LF migration.

FRET Revealed That PDGFRα and Nrp1 Enhance the Collagen Fiber–mediated Increase in Rac1-GTP

Earlier we demonstrated that soft collagen gels increase the abundance of Rac1-GTP in LF when assayed using a Rac1-activity biosensor (15). Therefore, we analyzed Rac1-FRET where control, PDGFRα-, or Nrp1-deleted LF contacted rigid fibers, to assess whether rigid collagen fibers localize Rac1-GTP within the cell. The Rac1-biosensor only exhibits FRET in its active GTP (not guanosine diphosphate [GDP])-bound form. Representative micrographs showing Rac1-FRET intensity under the various experimental conditions are shown in Figures 2A–2D. In LF obtained from TGCre$^{+/-}$ mice, with a single LoxP-flanked PDGFRα or Nrp1 allele (F/−), rigid collagen fibers increased Rac1-GTP. However, after Cre-mediated deletion, LF failed to exhibit increased Rac1-FRET when they contacted electrospun fibers (Figure 2E). Thus, when LF contact collagen fibers, PDGFRα and Nrp1 increase Rac1-GTP, which is an important signaling intermediate at the leading edge of migrating cells (24).

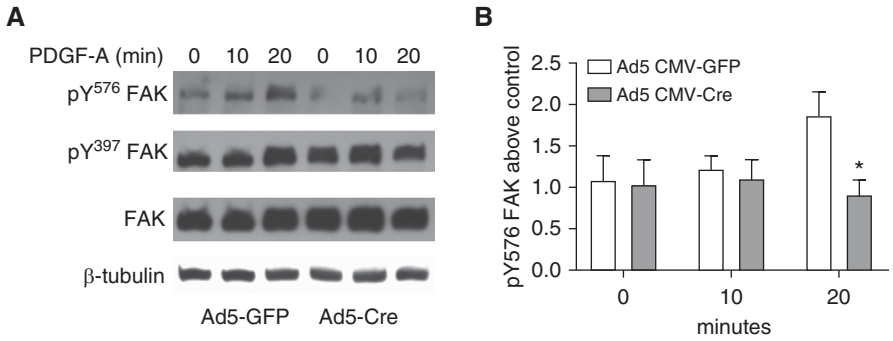

Figure 3. Nrp1 facilitates PDGF-A–mediated phosphorylation of FAK (focal adhesion kinase). Cultured LF from the lungs of Nrp1F/F mice that lacked *tagln*-Cre were transduced with an Adenovirus5 (Ad5) vector containing cytomegalovirus (CMV) promoter driving either eGFP (control) or Cre-recombinase (Cre) 40 hours before stimulation with 50 ng/ml PDGF-A. At the times shown, cell layers were harvested in buffer containing proteinase and phosphatase inhibitors and subjected to Western IB for pY^{397} or pY^{576} FAK. (*A*) Representative immunoblot. (*B*) The densitometric data at 20 minutes from five separate cell isolations were pooled, and LF from the two Ad5-CMV treatment conditions were compared within each isolation. $^{*}P < 0.05$, Student's *t* test, paired variables. eGFP = enhanced GFP.

Neuropilin-1 Enhances PDGF-A–induced Phosphorylation of Tyrosine 576 on FAK

During cell migration, PDGFRα phosphorylates Src, which promotes phosphorylation of FAK at tyrosine-576 (pY^{576}). Cultured LF from the lungs of mice bearing two Nrp1-Flox alleles were exposed to adenovirus Cre to deplete Nrp1. A representative Western immunoblot is shown in Figure 3A. We observed that PDGF-A increased phosphorylation of FAK-Y^{576} (which is completely dependent on Src-kinase) but not FAK-Y^{397}, which can also be autophosphorylated. However, this was not observed when Nrp1 was depleted (Figure 3B), indicating that Nrp1 enhances downstream signaling of PDGFRα through FAK (25, 26).

Nrp1 Does Not Alter the Abundance of Active TGFβ in Cultured LF

Cao and associates observed that that by regulating TGFβ signaling, Nrp1 was an autocrine regulator of PDGFRβ signaling and myofibroblast activation (27). We hypothesized that Nrp1 may alter the activation of latent TGFβ and thereby regulate the balance between the migratory synthetic and more stationary contractile phenotypes. We assessed whether Nrp1 deletion altered the activation of latent TGFβ, by coculturing control and Nrp1-deleted LF with MLE-Luc reporter cells. Control LF, which had been isolated from lungs of $TGCre^{+/-}$; Nrp1F/F mice, generated 5.82 ± 0.43 fmol, whereas those that had been isolated from $TGCre^{+/-}$; Nrp1F/− littermates generated 6.82 ± 1.08 fmol of active TGFβ per mg protein (not significant).

PDGFRα Regulates FA

During cell migration, FAK is phosphorylated at the leading edge, where Rac1-GTP is locally generated and FA are remodeled (28). Using immunofluorescence microscopy, we examined the accumulation of paxillin at FA in LF that were isolated from control, $TGCre^{+/-}$; PDGFRαF/−, or PDGFRα-depleted ($TGCre^{+/-}$; PDGFRαF/F) littermates and cultured on glass. Representative micrographs are shown in Figures 4A and 4B, with insets showing only phospho-paxillin (white) and paxillin (magenta). Depletion of PDGFRα did not reduce paxillin-containing pixels, when expressed as a proportion of the pixel area of the entire cell (Figure 4C). PDGFRα depletion reduced the proportion (relative to cell area) of F-actin–containing pixels (Figure 4D). Using an inhibitor of Rho-GTPase signaling, we observed that the areal density of actin-containing pixels decreased in control, but not in LF isolated from $TGCre^{+/-}$; PDGFRαF/F mice. These findings indicate that PDGFRα signaling regulates actin filament formation in a Rho-dependent fashion. When LF were embedded in a 100 μg/ml type I collagen gel, the mean area of individual paxillin-containing focal adhesions was higher in LF bearing the PDGFRα-deletion (Figure 4E).

β1-Integrin–Containing Clusters Are Larger When PDGFRα Signaling Is Reduced

β1 integrins are required for cells to bind fibrillar collagen and compose the extracellular portion of focal adhesions. Therefore, we examined the area of CD29 (integrin β1)-containing integrin clusters in LF isolated from $TGCre^{+/-}$; PDGFRαF/− and $TGCre^{+/-}$; PDGFRαF/F mice, which were cultured in three-dimensional (3D) collagen gels. The mean area of individual integrin clusters was larger in PDGFRα-depleted LF, although the combined area of all clusters within a cell did not change as a proportion of cell area (Figure 5A). This corroborates the observation that the mean area of individual, paxillin-containing FA was larger when PDGFRα was depleted. Lung fibroblasts from both genotypes were cultured in collagen gels containing electrospun collagen fibers, and the area of CD29 contacting electrospun fibers was compared. Representative micrographs are shown in Figure 5B. Although the areal density of CD29 was not reduced on a whole-cell basis (integrin clusters at focal adhesions as well as those overlying fibers), a lower fraction of fiber area contacted CD29 in PDGFRα-depleted LF (Figure 5C). Therefore, although CD29 accumulates at focal adhesions when PDGFRα signaling is reduced, less CD29 localizes over the more rigid electrospun collagen fibers, which may impair migration.

Analysis of LF-Surface–associated Collagen-Binding Integrins

The predominant fibrillar collagens in the lung are types I and III, which engage α1β1 or α2β1 integrin dimers on lung fibroblasts. We used flow cytometry to assess differences in collagen-binding integrins on fibroblasts with varied levels of PDGFRα or Nrp1. Representative dot plots are shown in Figure E4 to illustrate the gating strategies and population size. Using the PDGFRα-GFP knock-in, we observed that a larger proportion of PDGFRα-expressing (either GFP^{low} or GFP^{high}) than nonexpressing (GFP^{neg}) LF displayed CD49a (α1β1), expressed as a fraction of their respective parent (GFP^{high}, GFP^{low} or GFP^{neg}) population, when analyzed without culturing (Figure 6A). A smaller proportion of either GFP^{high} or GFP^{low} than GFP^{neg} LF displayed CD49b (α2β1). After culturing LF from mice

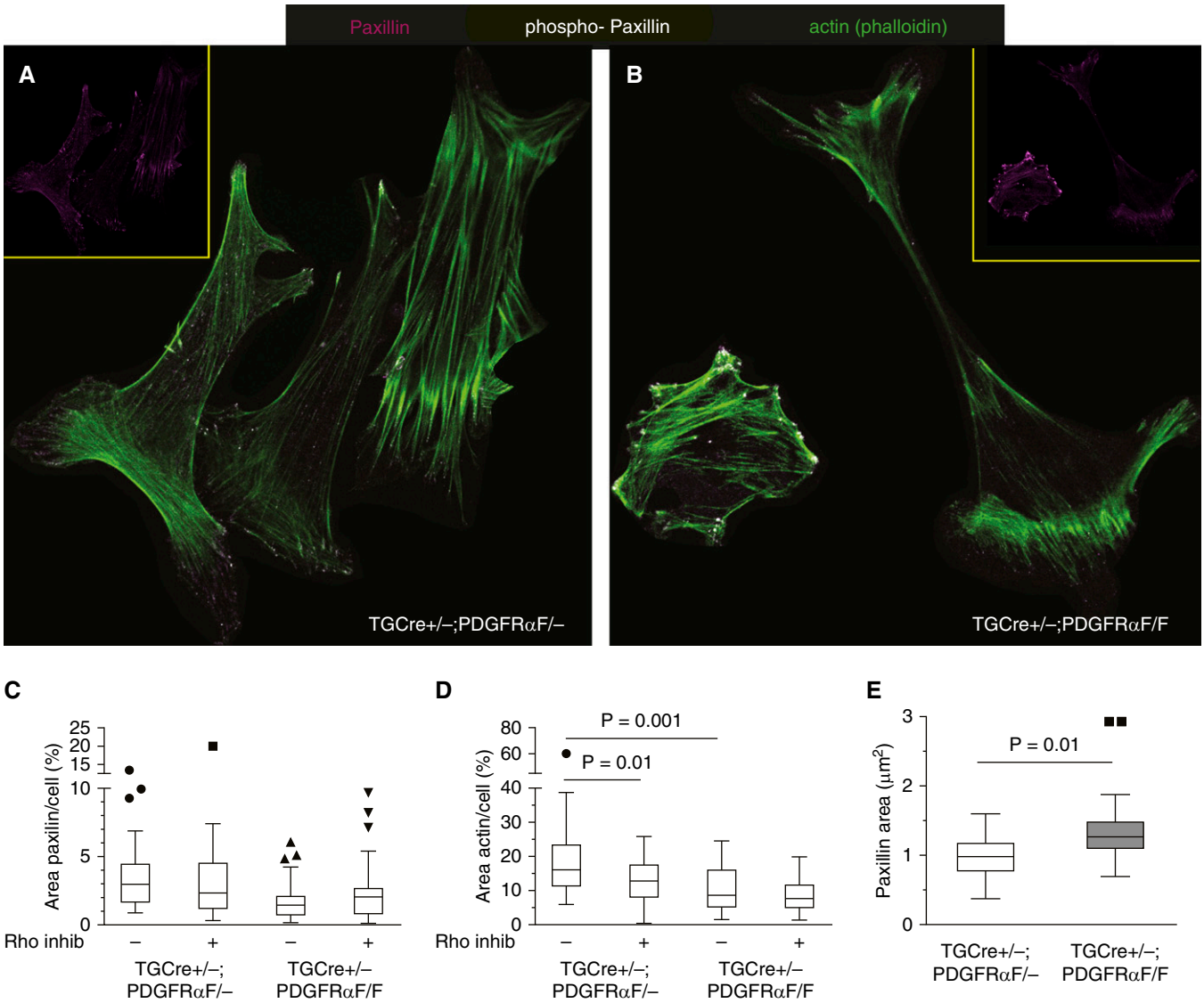

Figure 4. Targeted deletion of PDGFRα alters focal adhesions (FAs) in LF. LF were isolated from TGCre$^{+/-}$; PDGFRα F/− or TGCre$^{+/-}$; PDGFRαF/F mice and cultured on glass that had been coated with a thin layer of collagen (adsorbed from a 200 μg/ml solution in PBS). Before fixation, the cell layers were exposed to Rho inhibitor ectoenzyme C3 transferase (CT104) for 3 hours. Paxillin and phospho-paxillin in FA were stained and assigned magenta and white pseudo-colors, respectively, and F-actin fibers were stained with phalloidin (green). (*A* and *B*) Representative images acquired without exposure to Rho inhibitor. (*C* and *D*) Images were captured and analyzed using uniform thresholding, and the pixel areas occupied by paxillin (*C*), and actin (*D*) were normalized to the area occupied by the entire area of that particular cell. Box-and-whisker (Tukey) plots of pooled data from four separate cell isolations for each genotype, 40 to 60 cells per treatment condition, and analyzed using a two-way ANOVA. (*E*) LF were adhered to collagen-coated glass and then incubated in medium containing 100 μg/ml of collagen, which formed a gel around the cells. The area of paxillin in individual FAs was analyzed for 50 cells pooled from three separate cell isolations and compared using Student's *t* test for paired TGCre$^{+/-}$; PDGFRα F/− and TGCre$^{+/-}$; PDGFRαF/F littermates.

bearing the PDGFRα-GFP knock-in, the proportion of CD49b^{+} GFP^{pos} LF increased relative to the GFP^{neg} population (Figure 6B). This indicates that the proportions of LF displaying CD29 (β1 integrin) were not altered among LF with different levels of PDGFRα-GFP fluorescence or by culturing the cells. In contrast, the proportion of CD49b, although low in the freshly isolated PDGFRα-expressing cells, increased in culture.

We examined whether PDGFRα or Nrp1 deletion altered the activation state of collagen-binding β1 integrin. An antibody was used to identify LF displaying PDGFRα (CD140a) on their surface. The proportions of LF displaying CD29 (using an antibody that detects both active and inactive conformations) or CD49a were not altered in LF from TGCre$^{+/-}$; PDGFRαF/F or TGCre$^{+/-}$; Nrp1F/F mice compared with their littermate control mice (data not shown). Using an antibody that only detects CD29 in its active conformation, the proportion of active CD29 was higher in both CD140a^{neg} and CD140a^{pos} LF isolated

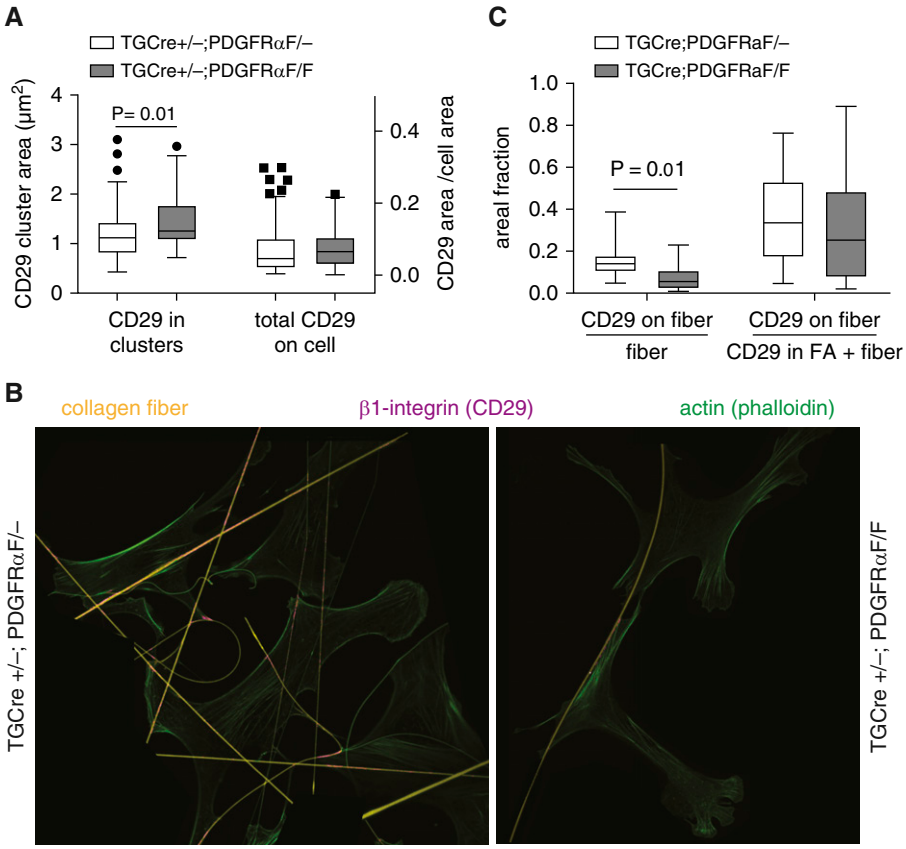

Figure 5. (*A–C*) β1-integrin–containing clusters in PDGFRα-depleted LF are larger, but fewer reside over collagen fibers. β1-integrin (CD29) clusters were visualized by immunostaining using cultured LF from TGCre$^{+/-}$; PDGFRα F/− and TGCre$^{+/-}$; PDGFRαF/F littermates in collagen gels: without collagen fibers (*A*) or with electrospun collagen fibers (*B* and *C*). (*B*) Fluorescein-collagen incorporated into the solution before electrospinning is pseudo-colored yellow, whereas phalloidin-stained actin filaments are green and CD29 is magenta. Uniform thresholding was used to quantify the mean area of CD29 within individual clusters (*A*, left) or the area per cell (right). CD29 localizing to pixels overlying fibers (CD29 on fiber) or clusters that did not colocalize with fibers (CD29 in FA) are shown in (*C*). Box-and-whisker (Tukey) plots of data for 70 cells for each genotype pooled from four separate cell isolations (*A*) and 25 cells for each genotype pooled from three separate cell isolations (*C*); paired *t* test comparing TGCre$^{+/-}$; PDGFRα F/− and TGCre$^{+/-}$; PDGFRαF/F littermates. Although individual clusters were larger in LF from TGCre$^{+/-}$; PDGFRαF/F mice, fewer localized over electrospun collagen fibers.

from TGCre$^{+/-}$; PDGFRαF/F mice than in those isolated from their TGCre$^{+/-}$; PDGFRαF/− littermates (Figure 6C). In contrast, the proportion of LF displaying active CD29 relative to the entire CD140a^{neg} population was lower for those isolated from TGCre$^{+/-}$; Nrp1F/F mice compared with their littermate control mice (Figure 6D). When combined with the observation that CD29-containing integrin clusters are larger in cultured LF from TGCre$^{+/-}$; PDGFRαF/F mice, this suggests that active CD29 may be retained on the plasma membrane when PDGFRα signaling is reduced.

PDGFRα Signaling Regulates the Positioning of Early Endosomes Containing CD29

The abundance and activation state of collagen-binding integrins is regulated by recycling through the very early and early endosomal compartments. We used the proximity ligation assay to examine the distribution CD29-containing early endosomes relative to the actin cytoskeleton. The formation and preservation of lamellipodia, where early endosomes may reside, is regulated by PDGFRα and Rho-GTPase. We did not observe (Figures E5A and E5B) a genotype-related difference in puncta showing colocalization of CD29 and EEA1 (early endosome antigen-1), in the absence or presence of an inhibitor of Rho signaling (Figure E5C). However, a larger proportion of the puncta were located in lamellipodia (distal to ventral actin fibers, Figure E5D). In LF with a normal complement of PDGFRα, inhibiting Rho signaling reduced the peripheral localization in LF, whereas it was not altered in LF isolated from TGCre$^{+/-}$; PDGFRαF/F mice (Figure E5D).

Collagen Fibers Are Abnormally Distributed in Lungs with Depleted PDGFRα

We have shown that lungs from TGCre$^{+/-}$; PDGFRαF/F have enlarged air spaces and fewer elastic fibers (21). Alveolar myofibroblasts, which contain abundant αSMA at P12, are reduced in these mice. We used SHG microscopy to determine whether collagen fibers are less abundant or aberrantly localized in lungs from TGCre$^{+/-}$; PDGFRαF/F mice. Lungs were uniformly inflated to 60% or 100% of total lung capacity and fixed. Representative micrographs from fully inflated lungs (Figure 7A) show that elastic fibers colocalize with αSMA 2–containing cells in lungs from both TGCre$^{+/-}$; PDGFRαF/− and TGCre$^{+/-}$; PDGFRαF/F mice, although they are reduced after PDGFRα deletion. As previously observed, αSMA was diminished and primarily limited to the alveolar ducts of TGCre$^{+/-}$; PDGFRαF/F mice, whereas it was more uniformly distributed in the secondary alveolar walls of TGCre$^{+/-}$; PDGFRαF/− mice (21). In their control littermates, collagen fibers colocalized with αSMA, whereas they were observed in locations that lacked αSMA in the lungs of TGCre$^{+/-}$; PDGFRαF/F mice. Likewise, collagen was observed in regions of the air spaces of PDGFRα-depleted lung, which lacked elastic fibers. For partially inflated lungs, histogram bins were created with upper limits corresponding to the pixel widths shown in Figure 7C. The collagen fiber width distribution was shifted toward larger fibers in TGCre$^{+/-}$; PDGFRαF/F mice. The genotype-related shift was not observed when the lungs were fully inflated (data not shown). We compared the angular distribution of collagen fibers relative to the air–tissue interface of the adjacent alveolar wall (Figure 7D). An angle of 0° represents a fiber that is parallel to the interface, whereas 90° is perpendicular. The data were binned into three angular ranges, to show that a smaller proportion of fibers were oriented near

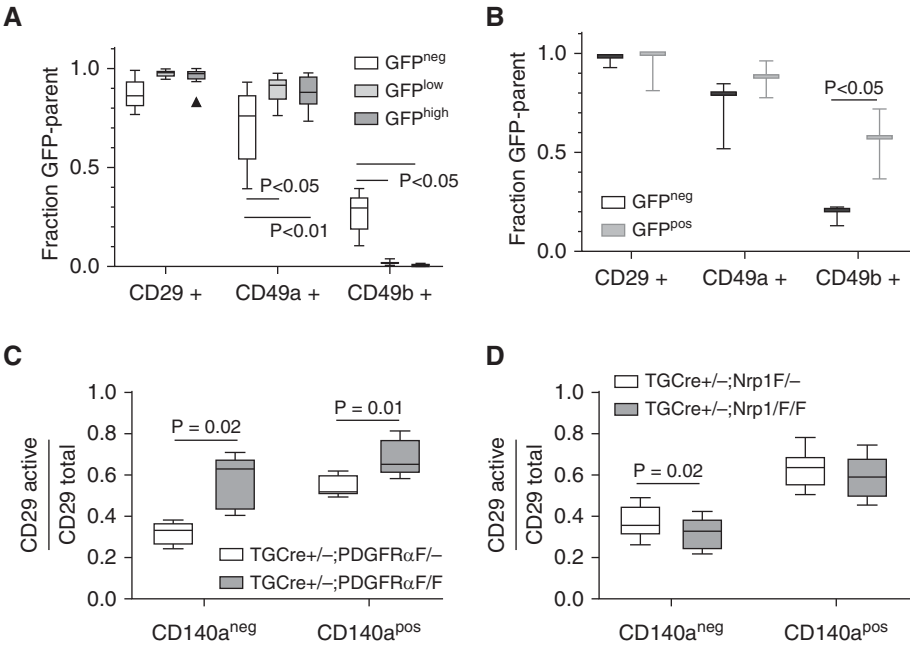

Figure 6. PDGFRα and Nrp1 modify the size of LF populations bearing specific collagen-binding integrins on their surface. Fibroblasts were isolated from the lungs of mice bearing the PDGFRα-GFP knock-in, or TGCre-targeted deletion of either PDGFRα ($TGCre^{+/-}$; PDGFRαF/F) or neuropilin-1 ($TGCre^{+/-}$; Nrp1F/F) and their respective litter control mice bearing only one LoxP-flanked allele (F−). Immediately after isolation (except in *B*), the LF were fixed and then subjected to immunostaining without using detergent to permeabilize the plasma membrane, therefore only identifying extracellular epitopes. The fluorescence resulting from specific isotype controls was subtracted from that observed using epitope-specific antibodies: CD140a, PDGFRα; CD29, β-integrin (CD29 total); CD49a, α1-integrin; CD49b, α2-integrin; or the epitope specific for activated CD29 (CD29 active). (*A* and *B*) Cells expressing PDGFRα were distinguished based on the fluorescence intensity of eGFP, for which transcription is regulated by the endogenous PDGFRα promoter. (*B*) LF were cultured before fixation and immunostaining. Box-and-whisker (Tukey) plots of data. *A* and *B* represents nine and three separate cell isolations, respectively, for each genotype for mice bearing the PDGFRα-GFP knock-in. *C* and *D* represent five and six separate cell isolations, respectively, for PDGFRα and Nrp1 deletions and their littermate control mice. A two-way ANOVA was used to analyze data in *A*, *C*, and *D*, and Student's *t* test for paired variables was used to analyze data for *B*.

parallel at partial inflation, whereas a larger proportion were oriented near perpendicular to the interface in lungs from $TGCre^{+/-}$; PDGFRαF/F mice.

Discussion

During pulmonary alveolar secondary septation, LF transit from the primary septa to reside in the walls of secondary septa, where collagen bundles may guide their relocation. We observed that LF with a full complement of PDGFRα and Nrp1 migrated more rapidly and exhibited greater Rac1 activation along rigid collagen fibers than their counterparts, which underwent targeted deletion of *pdgfrα* or *nrp1*. Mechanistically, LF with reduced PDGFRα formed larger FA (increased area of paxillin and CD29) when cultured in a 3D collagen gel. However, less CD29 localized along collagen fibers, suggesting that, unlike LF with a full complement of PDGFRα, fewer FA resided where the collagen substrate was more rigid. Correspondingly, we observed that MF did not uniformly colocalize with alveolar collagen fibers in mice bearing the *pdgfrα* deletion, whereas collagen colocalized with elastic fibers and αSMA in lungs of PDGFRα-sufficient mice. Deletion of *pdgfrα* did not alter the proportion of LF displaying CD29 on their surface, but a larger fraction of the CD29 was in the active configuration. Intracellular CD29 primarily colocalized with EEA1 in the perinuclear region of LF bearing the *pdgfrα* deletion, whereas a larger fraction was observed in the lamellipodia in LF without the deletion. These findings indicate that although FA form and integrins are activated when PDGFRα is reduced, LF aberrantly respond to contact with stiffer collagen fibers. Therefore, PDGFRα may condition signals from integrins and FA to optimize mechanotransduction and/or cytoskeletal remodeling.

Relevance of Our Culture Model to Alveolar Collagen Fibers

The size, orientation, and volume density of the electrospun collagen fibers in culture can be qualitatively compared with those in an alveolus. We observed (Figure 7) that collagen fibers are more abundant at the AER and align lengthwise parallel to the circumference, as others have observed using scanning EM (29). Electrospinning enables partial control of the orientation and density of collagen fibers, but without cellular boundaries they do not condense into bundles, as they do in the lung. Fibroblasts reorient collagen fibers in a surrounding gel to become more densely packed adjacent to the plasma membrane (30). The fibers gradually align parallel to the cell's long axis, promote polarization, and increase migration along this axis. Because electrospun fibers provided a preformed scaffold, we could assess the LF response to rigidity independent of their ability to reorient fibers.

The width of single hydrated electrospun fibers was 700 nm, approximating 1,000 nm for bundled collections (80–300 individual fibers) of collagen fibers in the AER (31). However, unlike the lung, the electrospun fibers were not prestressed, and their tension was static and not phasic, as during respiration. The volume density of collagen in the alveolar wall and AER is 0.03 and 0.12, respectively (31). The wall thickness of the alveolar ducts and alveoli is only 5 to 10 μm, which is thinner than the thickness required to form a stable gel. Although our collagen fiber/gel constructs were thicker than alveolar walls, LF formed intimate contacts with rigid fibers.

We used AFM to assess the elasticity of a composite containing electrospun collagen fibers (0.1 MPa) embedded in a collagen gel (∼0.1 kPa), which was similar to what LF encountered during migration (Figure 1). However, because the composite was fragile, we used a 2-mg/ml collagen gel bound to functionalize the glass, rather than the 1-mg/ml collagen gel on untreated glass, which we used for migration. Despite these modifications, we could not achieve

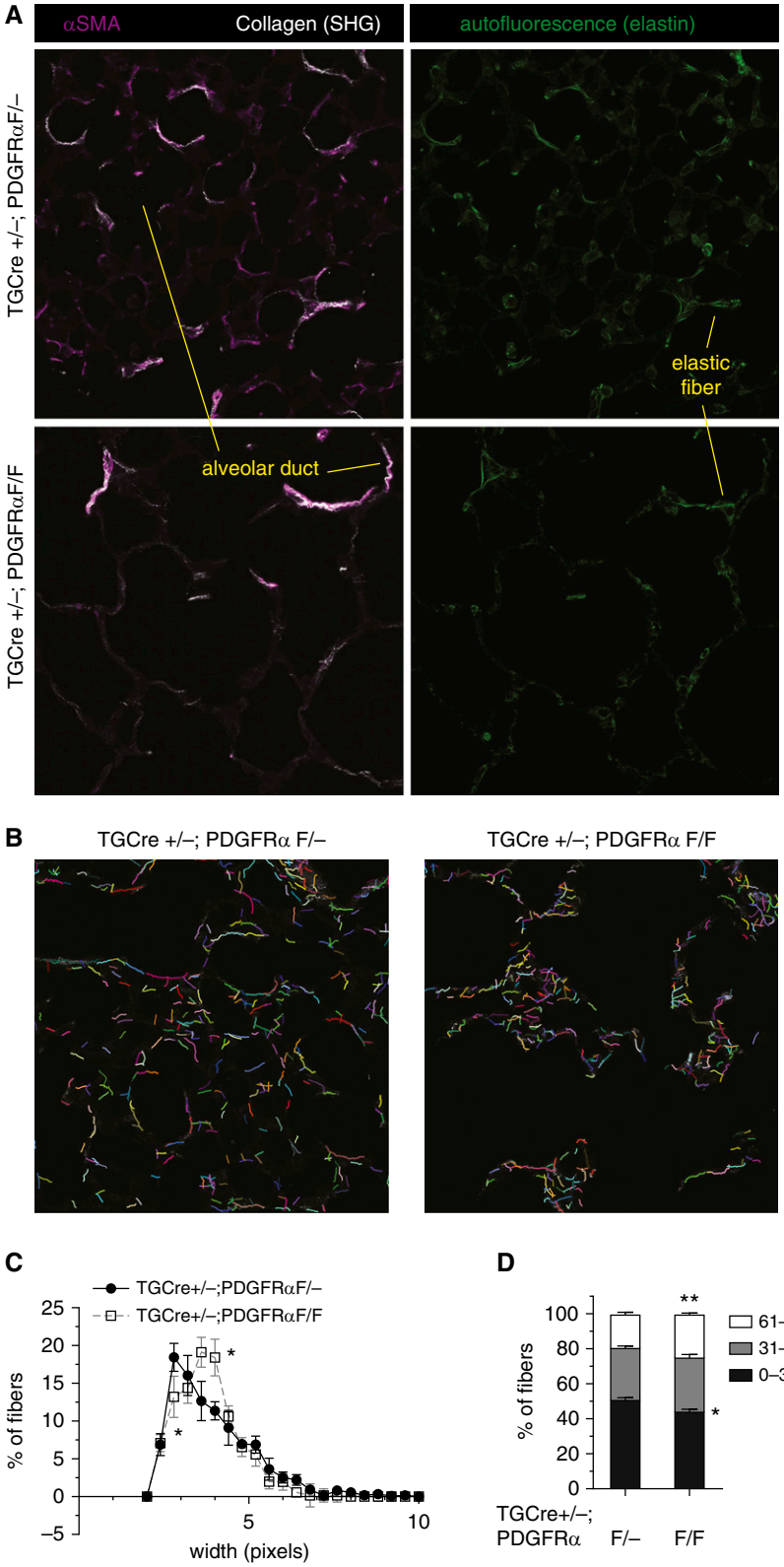

Figure 7. Collagen fibers are aberrantly distributed in lungs of mice bearing *tagln*-Cre–mediated PDGFRα gene deletion. On postnatal Day 12, lungs from TGCre$^{+/-}$; PDGFRαF/F and littermate control TGCre$^{+/-}$; PDGFRα F/− mice were uniformly inflated to 100% or 60% total lung capacity and fixed. Tissues were immunostained for ACTA2 (αSMA). Each imaging session comprised of a pair of lungs from control and PDGFRα littermates and, after optimization, the laser power and detector

direct contact between the cantilever tip and the fibers without engaging the cantilever arm. Thus, we were unable to probe individual fibers. Therefore, we can only conclude that the electrospun fibers stiffen the surrounding gel, in the proximity of LF. We observed that fibers alter the distribution of integrins on the cell surface and the speed of migration. Others observed that in the absence of cells, the stiffness of a 2-mg/ml collagen gel was ∼0.2 kPa (32), and fibroblasts increase the gel rigidity within a 20-μm radius by ∼10-fold (33). Therefore, if LF had been present, the composite may have been more rigid than we observed. The mean Young's modulus for the composite (0.185 ± 0.04 kPa) was lower than decellularized rat (0.38 ± 0.07 kPa) or human (1.6 ± 0.08) lung and native human lung (1.96 ± 0.13 kPa), measured using a similar cantilever tip (34, 35).

Collagen Fiber Rigidity Influences Migration and Rac Signaling by LF

Durotaxis, movement toward a more rigid substrate, requires spatially regulated dynamic formation and remodeling of FA. The "molecular clutch" model posits that integrin–cytoskeletal interactions surmount a force threshold, which induces retrograde flow of actin and engages the actin-Myo2 contractile unit to propel the cell forward (36). When the perceived forces are below this threshold, integrin-cytoskeletal linkages dissipate without engaging the cytoskeleton, so the "clutch" is not engaged. Conversely the "clutch" is disengaged when actin-myosin contractile forces overcome those sustained by an integrin-talin-actin interaction, and the FA linkage disbands. Protein tyrosine kinases modulate several of these interactions: *1*) the orientation and uniformity of FA with respect to the polarization axis, *2*) the force generated by and duration of actin-myosin contractions, and *3*) the stability of the linkage between the FA and actin filaments (37).

Signaling through PDGFRα and Nrp1 Regulate Focal Adhesions Overlying Collagen Fibers

When LF contacted and coaligned with electrospun collagen fibers, they generated more Rac1-GTP, and migrated faster along the fibers, than in their absence. But these responses were only observed when they possessed a full complement of PDGFRα and Nrp1. When PDGFRα was reduced,

the mean diameter of individual FA was larger (Figure 4), suggesting that the remodeling of FA requires PDGFRα. The specific contribution(s) of PDGFRα signaling to FA dynamics remain incompletely defined. Observing that PDGFRα-depleted LF maintained larger FA when contacting collagen fibers (Figure 5) suggests that PDGF-A signaling impacts the assembly or disassembly of larger integrin clusters. Because PDGFRα signals through Src and PI3K, diminished PTK or elevated phosphatase activity could enable nascent FA to enlarge by resisting dissolution.

It is less likely that diminished PDGFRα signaling enhances the assembly of mature FA. By phosphorylating paxillin, Src and FAK recruit vinculin and adapters to stabilize connections among integrins talin and actin (38). In contrast, we observed that PDGFRα depletion reduced the area occupied by actin, but not paxillin, in two-dimensional cultures (Figure 4). We also found that PDGF-A increased phosphorylation of Src (15) and FAK, which required Nrp1 (Figure 3). However, in most contexts, pSrc and pFAK increase phospho-paxillin (pPaxillin) when FA mature and enlarge (11). Our findings suggest that PDGFRα signaling does not enlarge FA by phosphorylation of Src and FAK. We could not directly analyze FA dynamics, because rapid sequence imaging with total internal reflection optics was unavailable. Overall, our findings suggest that PDGFRα signaling enables LF to adjust the size and organization of FA and to optimize cellular contacts with the more rigid electrospun fibers.

PDGFRα and Rac1-GTPase Regulate Signaling by Collagen through β1-Integrins and FA

In addition to altering the phosphorylation state of paxillin, Src and FAK regulate FA dynamics through Rac1. Others observed that Rac1-null mouse embryonic fibroblasts cannot form lamellipodia, although they spread and develop FA. The FA were larger, even though paxillin turnover was more rapid (39). We observed that CD29 clusters and the area of individual FA (paxillin immunostaining) were larger in LF after PDGFRα depletion (Figure 4). Our findings support earlier observations that although βPIX increased Rac1-GTP, it limited FA maturation and size (40).

We previously showed that the GEF (guanine exchange factor) GIT/PIX mediates activation of Rac1by PDGF-A in LF that were cultured in collagen (15). We have now shown that PDGF-A increases pY^{572}FAK, which can phosphorylate GIT/PIX and enhance its Rac1-GEF activity. PDGFRα depletion may reduce the phosphorylation of GIT/PIX and mitigate the stimulation of LF migration by Rac1 (41). Others have observed that βPix is particularly important for fibroblast migration through a 3D collagen matrix by limiting Rho-GTP (30). Phosphorylation of βPix by FAK directs Rac1 to focal adhesions and increases Rac1-GTP (13). Using Src and FAK biosensors that localize to focal adhesions, Wu and associates (12) showed that focal adhesions persist longer (disassembly was delayed) when FAK was inhibited. Observing that PDGFRα depletion (and PDGF-A signaling) correlates with larger focal adhesions is consistent with diminished disassembly. In summary, these findings support our hypothesis that collagen and PDGFRα cooperatively increase βPIX and Rac1 and limit FA maturation to promote more rapid LF migration.

PDGFRα and Rho-GTPase Regulate Actin Filaments and Extracellular Matrix Deposition

We observed that LF bearing a full complement of PDGFRα exhibited fewer actin stress fibers after inhibition of Rho-GTPase, whereas inhibition did not alter the area of fibers in PDGFRα-depleted LF (Figure 4D). Similarly, inhibition of Rho-activated kinase altered the positioning of CD29-containing early endosomes $TGCre^{+/-}$; PDGFRα F/−, but not $TGCre^{+/-}$; PDGFRαF/F LF (Figure E5D). Precisely how PDGFRα interacts with Rho-dependent signaling in lung fibroblasts remains unclear. By analogy to mechanosensitive, cancer-associated fibroblasts, myosin 2 is likely involved (42). Signaling through PDGFRα and Myo2, cancer-associated fibroblasts deposited parallel fibronectin fibers, which supported higher traction force and directed migration of epithelial cancer cells. Horikawa and associates showed that fibroblasts invading polyurethane sponges implanted in mice bearing a conditional deletion of PDGFRα produced fewer and less-ordered collagen fibers than sponges implanted in control mice (43).

PDGFRα Promotes an Orderly Distribution of Collagen Fibers

Using SHG optics to identify collagen fibers, we observed that when PDGFRα is preserved, collagen is a highly associated with αSMA-containing cells and aligned parallel to the boundary with the adjacent air space (Figure 7). In contrast, lungs from $TGCre^{+/-}$; PDGFRαF/F mice contained collagen fiber bundles where αSMA-containing LF were absent, and more of the fibers were aligned at angles approaching normal to the air space boundary. In partially but not fully inflated lungs, the width distribution of these fibers was shifted toward a larger diameter, suggesting that the fiber network structure changes with alveolar inflation. Our observations suggest that PDGFRα helps coordinate deposition of collagen and elastic fibers, so they function as a mechanical continuum (44).

These findings are relevant to lung mechanics at the organ level. Collagen fibers structurally intertwine and function in

Figure 7. (Continued). gains remained constant throughout the session. (*A*) Fully inflated lungs from both genotypes. The multi-photon second-harmonic generated (SHG) and confocal images from four partially inflated lungs for each genotype were analyzed using CT-Fire. (*B*) Tracings of reconstructed fibers using the curvelet function of CT-FIRE are shown. (*C*) CT-Fire was used to quantify the width of individual fibers, which were partitioned into 10 bins, and the number in each bin was expressed relative to the total enumerated fibers. *Multiple *t* tests (corrected for multiple comparisons), comparing the same bin for $TGCre^{+/-}$; PDGFRα F/− versus $TGCre^{+/-}$; PDGFRαF/F littermates. Images from four lungs of each genotype were used to quantify (Curve-Align algorithm) the angle of individual fibers relative to the most proximate point on the air–tissue interface. The angles were partitioned into three categories ranging from 0° to 30° (0° is parallel to the interface), 31°–60°, and 61°–90° (90° is perpendicular to the interface). $^{*}P < 0.05$ and $^{**}P < 0.01$, paired *t* test comparing littermates.

concert with elastic fibers, proteoglycans, and other fibrillar or matricellular proteins. The entire network is prestressed, and the mechanical forces encountered during deformation reflect the geometry of the network as well as the mechanical properties of each constituent. The contribution of collagen to the bulk elastic modulus varies with lung inflation because collagen is "wavy" at low lung volumes and tautens at high volumes. Our observation that fewer collagen fibers oriented parallel to the alveolar wall after *pdgfr*α deletion is consistent with our prior observation that this deletion diminished parenchymal elastance in adult TGCre$^{+/-}$; PDGFRαF/F mice (45). Although we focused on how collagen guides fibroblast migration, other extracellular matrix proteins likely also contribute. For example, fibronectin and α5β1 integrin are required for elastic fiber formation, and fibronectin enhances microfibril formation through fibrillin-1 (46, 47). And during alveolar fibrosis, fibronectin and αVβ1 integrin augment myofibroblasts, leading to more collagen deposition (48).

Our studies highlight how alveolar fibroblasts may respond to the elevated mechanical stresses at the AER during respiration. Although mechanical force fields may promote secondary septal formation, when distorted they may contribute to the progression of pulmonary emphysema. Progressive effacement of alveolar wall that accompanies emphysema leads to stress failure, which propagates to adjacent acini, leading to progression of emphysema long after cigarette smoking has stopped (49). Although collagen is deposited, it is disarrayed and fails to restore the balanced mechanical continuum that is established during developmental alveolar septation. Identifying how LF establish the mechanical continuum may uncover signaling pathways involving PDGFRα and Nrp1, which could be manipulated to help restore alveolar architecture and function in emphysema. ■

Author disclosures are available with the text of this article at www.atsjournals.org.

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